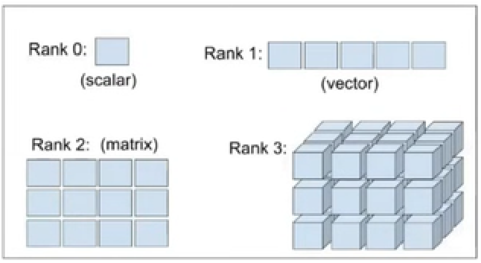
P2

Tensors

Tensor is specialized multi-dimensional array designed for mathematical and computational efficiency.



Matrix Operation

```
f = torch.randint(size=(2, 3), low=0, high=10)
g = torch.randint(size=(3, 2), low=0, high=10)

torch.matmul(f, g)

torch.dot(f, g)

torch.transpose(f, 0, 1)

h = torch.randint(size=(3, 3),low=0,high=10
,dtype=torch.float32)

torch.det(h)

torch.inverse(h)
```

Comparison Operations

```
i = torch.randint(size=(2, 3), low=0, high=10)
j = torch.randint(size=(2, 3), low=0, high=10)

i > j

i < j

i == j

i != j
```

Special Functions

```
k = torch.randint(size=(2, 3),high=10)

torch.log(k)

torch.exp(k)

torch.sqrt(k)

torch.softmax(k.float(), dim=0)

torch.relu(k)

torch.sigmoid(k)
```

Inplace Operations

```
m = torch.rand(2, 3)
n = torch.rand(2, 3)

m + n

id(m)
id(n)

id(m +n)

m.add_(n)

m.relu_()
```

Coping a Tensor

```
q = torch.rand(2, 3)

b = a

a[0][0] =  0

c = torch.clone(a)
```

https://github.com/Rudra-G-23/deep-learning-using-pytorch